

**Detecting and Removing PFAS from
Drinking Water in Developing Countries:
Practical and Affordable Strategies**

ENGW 3315

Northeastern University

08.07.2026

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Abstract

Per- and polyfluoroalkyl substances (PFAS) are persistent synthetic chemicals that have become a growing environmental and public health concern because of their widespread use and resistance to degradation. Although reliable methods for detecting and removing PFAS have been developed, many remain too expensive or resource-intensive for widespread implementation in developing countries. This literature review examines current research on PFAS detection, treatment, and regulation through three themes: the accessibility of detection methods, the tradeoffs between treatment effectiveness and affordability, and the role of environmental policy and infrastructure in successful PFAS management. The reviewed studies consistently identify LC-MS/MS as the analytical standard for PFAS detection and recognize several effective treatment technologies, but they also emphasize that cost, technical expertise, and limited regulatory capacity restrict their implementation. Overall, the literature suggests that improving PFAS management in developing countries depends less on developing new technologies than on making existing detection and treatment methods more affordable, practical, and accessible while strengthening long-term monitoring and environmental governance.

Introduction

Per- and polyfluoroalkyl substances (PFAS), often called "forever chemicals," are a large group of synthetic compounds widely used in industrial processes and consumer products because of their resistance to heat, water, and chemical degradation. These same properties allow them to persist in the environment, contaminate drinking water, and accumulate in living

organisms, raising concerns about their potential effects on both human health and ecosystems. While PFAS contamination has been extensively studied in developed countries, many developing nations face additional challenges due to limited monitoring capacity, weaker regulatory frameworks, and financial constraints that restrict their ability to detect and remove these contaminants. Although researchers generally agree on the most effective methods for PFAS detection and treatment, these technologies are often too expensive or resource-intensive for widespread implementation in developing countries¹⁻⁶. This literature review examines current research on PFAS detection, removal, and regulation to evaluate which strategies are both scientifically reliable and realistically achievable in resource-limited settings. It argues that the biggest issue is making existing detection and treatment methods affordable, practical, and accessible enough for long-term use.

Detecting PFAS in Developing Countries: Reliability versus Affordability

Before any removal or treatment of PFAS can occur, detection must first occur. Only when something can be measured can it be reliably monitored and treated. Another factor that must be considered is the constraints developing countries have toward implementing reliable PFAS detection. Budget becomes a much larger factor, along with having enough professionals with experience to perform PFAS analysis. The initial cost of instrumentation that is not readily available in many of these countries must also be considered, as the specialized equipment required for PFAS detection is often prohibitively expensive.

All of the sources that talk about detection agree that LC-MS/MS, typically paired with solid-phase extraction (SPE), is currently the most reliable method for PFAS detection^{1,3,5,6}, but they also acknowledge that it is expensive and requires specialized equipment and technical

expertise that developing countries may struggle to acquire and maintain. Adewuyi and Li emphasize accessibility as the primary limitation to implementing these methods, while Umejuru et al. identify weak laboratory infrastructure, limited funding, and weaker governance as additional barriers that prevent widespread monitoring across Africa^{1,5}. Both sources identify these issues but do not offer many practical solutions beyond recognizing the need for greater investment and monitoring capacity.

Moosa et al. take the discussion a step further by proposing possible alternatives to traditional laboratory methods, including biosensors, infrared spectroscopy, and electrochemical detection techniques³. Rather than replacing LC-MS/MS entirely, the authors argue that future detection methods should focus on becoming more portable, affordable, and practical while still maintaining acceptable analytical performance³. This makes Moosa et al. one of the few sources that moves beyond identifying the problem and instead suggests directions for future research³. Yamazaki et al. further reinforce the importance of reliable monitoring by demonstrating how standardized analytical methods produced detailed evidence of changing PFAS contamination in India's Cooum River following the 2015 Chennai flood⁶. While Yamazaki et al. do not focus on affordability, the study highlights why accurate monitoring remains essential for understanding contamination trends and supporting future regulation⁶. Together, these studies suggest that the greatest challenge is no longer determining the best analytical method, but making reliable PFAS detection accessible enough for routine use in developing countries.

Removing PFAS from Drinking Water: Effectiveness versus Cost

Detecting PFAS is only the first step in addressing contamination. Once PFAS have been identified, the next challenge becomes selecting treatment methods that are both effective and

economically feasible for long-term use. Adewuyi and Li, Moosa et al., and Umejuru et al. all discuss PFAS removal, and while they generally agree on the most promising treatment technologies, they also emphasize the tradeoffs involved in implementing them in developing countries^{1,3,5}. Activated carbon, ion exchange resins, membrane filtration, and advanced oxidation or photoelectrocatalytic processes are consistently identified as the leading treatment methods, but each differs in removal efficiency, operating cost, maintenance requirements, and long-term sustainability^{1,3,5}. Another recurring concern throughout these studies is that many treatment methods do not completely destroy PFAS but instead transfer them from water into another waste stream that must still be managed. This challenge becomes even greater for short-chain PFAS, which are becoming increasingly common and are generally more difficult to remove than their long-chain counterparts^{1,3,5}.

Adewuyi and Li and Moosa et al. both identify adsorption using activated carbon, ion exchange resins, and membrane filtration as the most established methods currently available for PFAS removal^{1,3}. Rather than disagreeing over which technology performs best, the authors emphasize that each offers advantages under different conditions^{1,3}. Activated carbon remains one of the most widely used treatment methods because of its relative simplicity and effectiveness for many long-chain PFAS, while ion exchange resins generally provide higher removal efficiencies but require greater operational costs^{1,3}. Membrane filtration techniques such as reverse osmosis are capable of removing a broad range of PFAS compounds, although they demand substantial energy, specialized equipment, and ongoing maintenance^{1,3}. Together, these studies suggest that selecting a treatment technology depends not only on removal performance but also on whether the system can realistically be installed and maintained in resource-limited settings.

Although these technologies are capable of removing PFAS from drinking water, the sources consistently point out that removal does not necessarily eliminate the contamination. Adewuyi and Li explain that adsorption and membrane-based systems often concentrate PFAS into spent filter media or waste brine rather than destroying the compounds themselves¹. Moosa et al. expand upon this concern by reviewing emerging destruction technologies, including advanced oxidation and photoelectrocatalytic degradation, which attempt to permanently break down PFAS instead of transferring them into another waste stream³. While these approaches represent an important direction for future research, the authors acknowledge that many remain expensive, technically complex, and difficult to scale for widespread use. As a result, current treatment technologies frequently exchange one environmental challenge for another by creating concentrated PFAS waste that still requires careful disposal.

Another point of agreement across the reviewed studies is the growing challenge presented by short-chain PFAS. Adewuyi and Li note that many conventional treatment technologies become less effective as industries transition from long-chain compounds toward shorter-chain alternatives¹. Yamazaki et al. indirectly reinforce this trend by documenting a noticeable increase in short-chain PFAS within sediment samples collected after the 2015 Chennai flood, suggesting that these compounds are becoming increasingly prevalent in developing countries⁶. Rather than viewing this as an isolated finding, Moosa et al. argue that future treatment strategies must account for these changing contamination patterns while remaining practical enough for routine implementation. Together, these studies indicate that future research should focus not only on improving removal efficiency but also on developing treatment technologies capable of effectively destroying short-chain PFAS without creating additional environmental burdens. Rather than searching for a single universal solution, the

literature increasingly supports balancing effectiveness, affordability, and long-term sustainability when selecting PFAS treatment strategies for developing countries.

Regulation and Infrastructure

Even the most effective methods for detecting and removing PFAS cannot reduce contamination if governments lack the policies and infrastructure needed to implement them. Sharma et al., Umejuru et al., Adewuyi and Li, and Anik et al. all emphasize that regulation plays a major role in determining whether PFAS management strategies are successful^{1,2,4,5}. Many developing countries face limited environmental monitoring, inconsistent enforcement, and financial constraints that make adopting PFAS detection and treatment technologies much more difficult. At the same time, differences in international regulations create opportunities for PFAS production and pollution to shift toward countries with weaker environmental protections. Rather than viewing regulation as separate from detection and treatment, these studies suggest that effective PFAS management depends on combining scientific advances with stronger monitoring systems, improved governance, and coordinated environmental policies.

Sharma et al. place the greatest emphasis on the role of policy by arguing that unequal PFAS regulations between developed and developing countries contribute directly to the continued spread of contamination⁴. As wealthier nations adopt stricter environmental standards, PFAS production and manufacturing can shift toward countries with weaker regulations and lower operating costs⁴. The authors describe this process as a "race to the bottom," where developing countries bear a disproportionate share of the environmental and public health burden while lacking the resources needed to monitor or manage contamination effectively⁴. Rather than viewing PFAS as a problem that individual countries can solve independently, Sharma et al.

argue that international cooperation and more consistent regulatory standards are necessary to prevent pollution from simply moving across national borders⁴.

Although Sharma et al. examine the issue from a global policy perspective, Umejuru et al. demonstrate how these regulatory challenges appear at the regional level in Africa⁵. Their review found that PFAS monitoring has been conducted in only a limited number of African countries, leaving large portions of the continent without reliable data on contamination levels⁵. The authors attribute these gaps not only to limited laboratory capacity but also to insufficient funding, weak environmental governance, and inconsistent enforcement of existing regulations⁵. Similarly, Adewuyi and Li argue that many developing countries continue to struggle with monitoring because environmental regulations have developed more slowly than industrial PFAS use¹. Anik et al. reach a comparable conclusion by showing that although PFAS contamination has been documented across many developing countries, the available data remain fragmented and concentrated in only a small number of regions². Together, these studies suggest that limited scientific information is often a consequence of limited institutional capacity rather than an absence of contamination itself.

Yamazaki et al. provide an example of why stronger monitoring and regulation remain important even after contamination has been identified. By comparing sediment collected before and after the 2015 Chennai flood, the authors documented a substantial increase in PFAS concentrations and a shift toward shorter-chain compounds that had previously gone undetected⁶. Their findings suggest that environmental conditions and continued industrial activity can rapidly change PFAS contamination patterns, reinforcing the need for consistent long-term monitoring rather than isolated sampling efforts⁶. Although the study focuses on a single river

system, it demonstrates how localized monitoring can provide evidence that supports broader environmental policy and future regulatory decisions⁶.

Taken together, these studies suggest that advances in detection and treatment technologies alone will not solve PFAS contamination in developing countries. Scientific progress has produced reliable analytical methods and increasingly effective treatment technologies, but these tools cannot be fully utilized without the regulatory frameworks, funding, and institutional capacity needed to support them. The literature consistently indicates that successful PFAS management depends on integrating scientific research with stronger environmental governance, expanded monitoring programs, and greater international cooperation. Without these broader structural improvements, even the most effective technologies will have a limited impact on reducing PFAS contamination over the long term.

Conclusion

The studies reviewed consistently demonstrate that the scientific understanding of PFAS detection and treatment has advanced considerably, yet significant challenges remain for developing countries¹⁻⁶. Researchers largely agree on reliable analytical techniques and identify several effective treatment technologies, but these methods often require financial resources, technical expertise, and infrastructure that are unavailable in many resource-limited settings. Rather than disagreeing about the science itself, the reviewed sources differ primarily in the aspects of the problem they emphasize. Adewuyi and Li, Anik et al., and Umejuru et al. focus on documenting the extent of PFAS contamination and the barriers to monitoring it, Moosa et al. evaluate both established and emerging detection and treatment technologies, Sharma et al. examine the role of international regulation and environmental policy, and Yamazaki et al.

provide direct evidence of how contamination patterns continue to evolve in a developing country¹⁻⁶. Considered together, these perspectives provide a more complete understanding of the issue than any individual study alone by connecting scientific, economic, and policy challenges into a single conversation.

Although important progress has been made, the literature identifies several questions that remain unresolved³. Current treatment technologies often transfer PFAS into another waste stream rather than permanently destroying the compounds, and short-chain PFAS continue to present growing challenges for both detection and removal. Likewise, many promising alternatives to conventional analytical methods remain too experimental or insufficiently validated for routine implementation³. Future research should therefore focus on developing affordable and portable detection technologies, improving treatment methods capable of permanently degrading short-chain PFAS, and strengthening the monitoring capacity and regulatory frameworks needed to support long-term implementation. Ultimately, the greatest barrier to improving PFAS management in developing countries is not the absence of scientific knowledge, but ensuring that existing technologies become practical, accessible, and economically feasible for the communities that need them most.

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